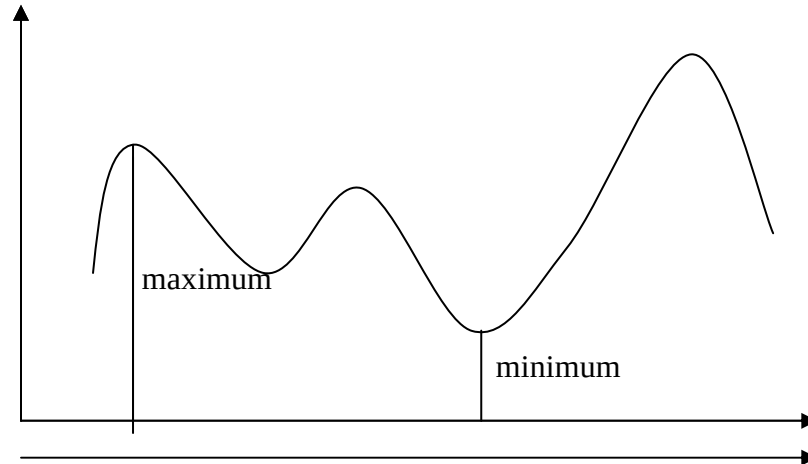


Maxima and Minima

Definition: A function $f(x)$ is said to have a maximum for a value a of x if $f(a)$ is greater than any other value that the function can have in the small neighbourhood of $x=a$.

Similarly, a function $f(x)$ is said to have a minimum for a value a of x if $f(a)$ is less than any other value that the function can have in the small neighbourhood of $x=a$.



1

Taylor's Theorem:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \dots$$

$$f(a-h) = f(a) - hf'(a) + \frac{h^2}{2!} f''(a) - \frac{h^3}{3!} f'''(a) + \dots$$

This gives

$$f(a+h) - f(a) = hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \dots$$

$$f(a-h) - f(a) = -hf'(a) + \frac{h^2}{2!} f''(a) - \frac{h^3}{3!} f'''(a) + \dots$$

Now considering $f'(a)$ only,

$$f(a+h) - f(a) = hf'(a)$$

$$f(a-h) - f(a) = -hf'(a)$$

Since for maximum $f(a+h) - f(a) < 0$ and minimum $f(a-h) - f(a) > 0$, from above two equations, we conclude that maximum and minimum value is possible only when $f'(a) = 0$.

Now considering $f''(a)$ only we have

$$f(a+h) - f(a) = \frac{h^2}{2!} f''(a)$$

$$f(a-h) - f(a) = \frac{h^2}{2!} f''(a)$$

Since for maximum value $f(a+h) - f(a) < 0$ and also $f(a-h) - f(a) < 0$, we conclude that when $f''(a) < 0$ we have a maximum value at $x = a$.

Since for minimum value $f(a+h) - f(a) > 0$ and also $f(a-h) - f(a) > 0$, we conclude that when $f''(a) > 0$ we have a minimum value at $x = a$.

If $f''(a) = 0$, we consider $f'''(a)$ only, then

$$f(a+h) - f(a) = \frac{h^3}{3!} f'''(a)$$

$$f(a-h) - f(a) = -\frac{h^3}{3!} f'''(a)$$

Since for maximum $f(a+h) - f(a) < 0$ and minimum $f(a+h) - f(a) > 0$, from above two equations, we conclude that, if $f'''(a) \neq 0$ then neither maximum nor minimum value

If $f'''(a) = 0$, we consider $f^{iv}(a)$ only

$$f(a+h) - f(a) = \frac{h^4}{4!} f^{iv}(a)$$

$$f(a-h) - f(a) = \frac{h^4}{4!} f^{iv}(a)$$

Since for maximum value $f(a+h) - f(a) < 0$ and also $f(a-h) - f(a) < 0$, we conclude that when $f^{iv}(a) < 0$ we have a maximum value at $x = a$.

Since for minimum value $f(a+h) - f(a) > 0$ and also $f(a-h) - f(a) > 0$, we conclude that when $f^{iv}(a) > 0$ we have a minimum value at $x = a$.

And so on.

Working method:

If function $f(x)$ be given, find $f'(x)$ and equate it to zero. Solve this equation for x .

Let its roots be $a_1, a_2, a_3, \dots$

Find $f''(x)$ and hence find $f''(a_1), f''(a_2), \dots$

If $f''(a_1)$ is negative we have a maximum at $x = a_1$.

If $f''(a_1)$ is positive we have a minimum at $x = a_1$.

If $f''(a_1) = 0$ find $f'''(x)$ and then $f'''(a_1)$.

If $f'''(a_1) \neq 0$, then there is neither maxima nor minima at $x = a_1$.

If $f'''(a_1) = 0$; find $f^{iv}(x)$ and then $f^{iv}(a_1)$.

If $f^{iv}(a_1)$ is negative, then $f(x)$ is maximum and if $f^{iv}(a_1)$ is positive, then $f(x)$ is minimum at $x = a_1$.

If $f^{iv}(a_1) = 0$ Then find $f^v(a_1)$ and so on.

$$f'(a_1) = f''(a_1) = \dots = f^{n-1}(a_1) = 0 \text{ and } f^n(a_1) \neq 0$$

(1) if n be odd, then there is neither maxima nor minima at $x = a_1$

(2) if n be even and if $f^n(a_1)$ is negative then $f(x)$ is maximum at $x = a_1$ and

(3) if $f^n(a_1)$ is positive then $f(x)$ is minimum at $x = a_1$.

Problem-1: Find the maximum and minimum values of $y = x^5 - 5x^4 + 5x^3 - 10$.

Solution:

Let $f(x) = y = x^5 - 5x^4 + 5x^3 - 10$

$$f'(x) = 5x^4 - 20x^3 + 15x^2$$

$$f''(x) = 20x^3 - 60x^2 + 30x$$

For maximum and minimum, $f'(x) = 0$

$$\Rightarrow 5x^4 - 20x^3 + 15x^2 = 0$$

$$\Rightarrow x^4 - 4x^3 + 3x^2 = 0$$

$$\Rightarrow x^2(x^2 - 4x + 3) = 0$$

$$\Rightarrow x^2(x-3)(x-1) = 0$$

$$\therefore x = 0, 1, 3$$

When $x = 1$, $f''(1) = 20 - 60 + 30 = -10 < 0$

Therefore, the given function is maximum when $x = 1$

and maximum value is $f(1) = 1 - 5 + 5 - 10 = -9$

When $x = 3$, $f''(3) = 540 - 540 + 90 = 90 > 0$

Therefore, the given function is minimum when $x = 3$

and the minimum value is $f(3) = 243 - 405 + 135 - 10 = -37$

When $x = 0$, $f''(0) = 0$

Therefore the test fails.

$$\therefore f'''(x) = 60x^2 - 120x + 30$$

When $x = 0$, $f'''(0) = 30 \neq 0$

Therefore, the given function is neither maximum nor minimum when $x = 0$

Problem-2: Find the maximum and minimum values for $f(x) = 3x^5 - 25x^2 + 60x$.

Problem-3: Find the maximum and minimum values for $f(x) = x^6 - 12x^5 + 36x^4 + 4$.

Problem-4: Find the extremum values of $\frac{x^3}{3} + ax^2 - 3a^2x$.

Solution:

Let $f(x) = \frac{x^3}{3} + ax^2 - 3a^2x$

$$f'(x) = \frac{3x^2}{3} + 2ax - 3a^2$$

$$\Rightarrow f'(x) = x^2 + 2ax - 3a^2$$

$$f''(x) = 2x + 2a$$

For maximum and minimum, $f'(x) = 0$

$$\Rightarrow x^2 + 2ax - 3a^2 = 0$$

$$\Rightarrow x^2 + 3ax - ax - 3a^2 = 0$$

$$\Rightarrow x(x + 3a) - a(x + 3a) = 0$$

$$\Rightarrow (x + 3a)(x - a) = 0$$

$$\therefore x = a, -3a$$

When $x = -3a$, $f''(-3a) = -6a + 2a = -4a < 0$

Therefore, the given function is maximum when $x = -3a$
and maximum value is $f(-3a) = 9a^3$

When $x = a$, $f''(a) = 2a + 2a = 4a > 0$

Therefore, the given function is minimum when $x = a$

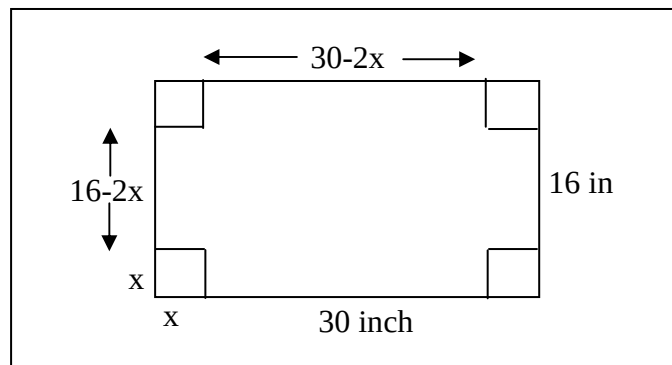
and minimum value is $f(a) = \frac{5}{3} a^3$

Problem-5: Find the maximum and minimum values for $f(x) = 1 + 2 \sin x + 3 \cos^2 x$.

Problem-6: Find the maximum and minimum values for $f(x) = \left(\frac{\log x}{x} \right)$.

Problem-7: Find the maximum and minimum values for $f(x) = x^{1/x}$.

Problem-8: An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume?



Let x be the length of the square to be cut out and V be the volume of the resulting box. Because we are removing a square of side x from each corner, the resulting box will have dimensions $16 - 2x$ by $30 - 2x$ by x .

Since the volume of a box is the product of its dimensions, we have

$$\begin{aligned} V &= (16 - 2x)(30 - 2x)x \\ &= 4x(8 - x)(15 - x) \\ &= 480x - 92x^2 + 4x^3 \end{aligned}$$

Because x represents a length, it cannot be negative and because the width of the cardboard is 16 inches, we cannot cut out squares whose sides are more than 8 inches long. Thus the variable x must satisfy $0 \leq x \leq 8$.

$$\begin{aligned} \text{Now } \frac{dV}{dx} &= 480 - 184x + 12x^2 \\ &= 4(x - 12)(3x - 10) \end{aligned}$$

$$\begin{aligned} \text{Setting } \frac{dV}{dx} = 0 \text{ yields } 4(x - 12)(3x - 10) &= 0 \\ \Rightarrow (x - 12)(3x - 10) &= 0 \end{aligned}$$

$$\therefore x=12 \text{ and } x = \frac{10}{3}$$

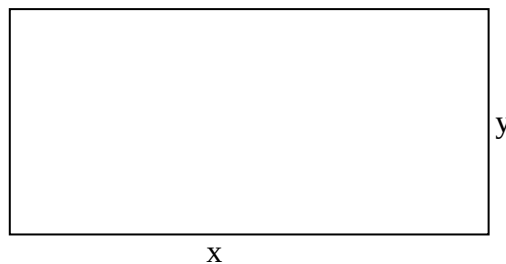
Since $x=12$ falls outside the interval $[0, 8]$ the maximum value of V occurs either at the critical number $x = \frac{10}{3}$ or at the endpoints $x=0$, $x=8$.

x	0	$\frac{10}{3}$	8
$V = (16 - 2x)(30 - 2x)x$	0	$\frac{19600}{27}$	0

Therefore maximum volume occurs when $x = \frac{10}{3}$ inches and maximum volume is $\frac{19600}{27}$.

Problem-9: A garden is to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 100 running feet of chicken wire is available for the fence?

Solution:



Let x (ft) be the length, y (ft) be the width and A (ft^2) be the area of the rectangular garden. Then $A = xy$ (i)

Since the perimeter of the rectangular garden is 100 ft., the variables x and y are related by the equation

$$2x + 2y = 100$$

$$\text{or, } y = 50 - x \text{ (ii)}$$

Substituting (ii) in (i)

$$A = x(50 - x) = 50x - x^2 \text{ (iii)}$$

Since x represents the length it can't be negative and since the two sides of length x cannot have a combined length exceeding the total perimeter of 100 ft, the variable x must satisfy $0 \leq x \leq 50$.

Differentiating (iii) with respect to x we have,

$$\frac{dA}{dx} = 50 - 2x$$

$$\text{Setting } \frac{dA}{dx} = 0 \text{ we obtain}$$

$$50 - 2x = 0$$

Or, $x=25$

Thus the maximum occurs at one of the values $x=0$, $x=25$, $x=50$.

Substituting these values in (iii) we get

x	0	25	50
$A=x(50-x)$	0	625	0

$\therefore$ The maximum area is 625 ft^2 occurs at $x=25$

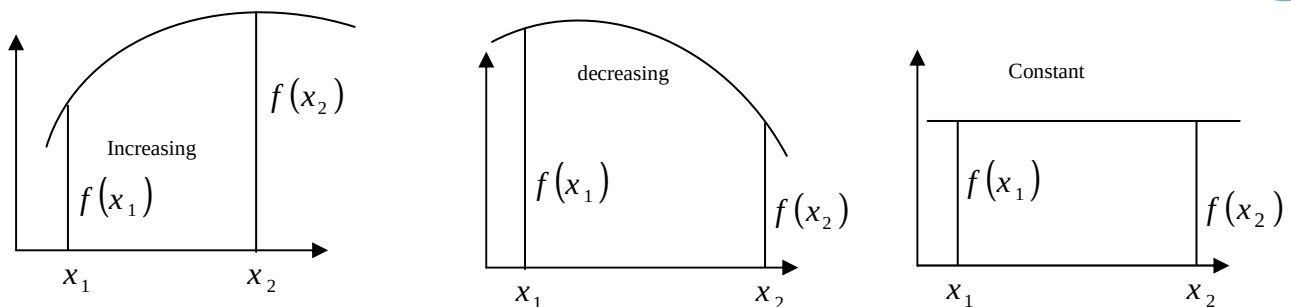
Substituting $x=25$ in (ii) $y=50-25=25$

So the rectangle of perimeter 100 ft with greater area is a square with sides of length 25 ft.

Analysis of Functions: Increasing, Decreasing, Concave up and Concave down

Definition: Let f be defined on an interval and let x_1 and x_2 denote numbers in that interval.

- (a) f is increasing on the interval if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (b) f is decreasing on the interval if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (c) f is constant on the interval if $f(x_1) = f(x_2)$ for all x_1 and x_2 .



Theorem: Let f be a function that is continuous on $[a, b]$ and differentiable on (a, b) then

- (a) If $f'(x) > 0 \forall x \in (a, b)$ then f is increasing on $[a, b]$.
- (b) If $f'(x) < 0 \forall x \in (a, b)$ then f is decreasing on $[a, b]$.
- (c) If $f'(x) = 0 \forall x \in (a, b)$ then f is constant on $[a, b]$.

Definition: If f is differentiable on an open interval I , then f is said to be concave up on I if f' is increasing on I and f is said to be concave down on I if f' is decreasing on I .

Theorem: If f be twice differentiable on an open interval I .

- (a) if $f'' > 0$ on I , then f is concave up on I .
- (b) if $f'' < 0$ on I , then f is concave down on I .

Definition: Inflection Point:

If f is continuous on an open interval containing x_0 and if f changes the direction of its concavity at the point $(x_0, f(x_0))$ then we say that f has an inflection point at x_0 and we call the point $(x_0, f(x_0))$ on the graph of f an inflection point of f .

Critical value:

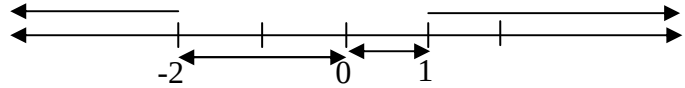
Values of the domain of f at which either $f'(x) = 0$ or f is not differentiable are called Critical Values / numbers.

Problem-1: Find the critical points of $y = 3x^4 + 4x^3 - 12x^2 + 2$. Identify the intervals on which y is increasing and decreasing, concave up and concave down. Also find the inflection points, if any, of the above function.

Solution:

Here $y = 3x^4 + 4x^3 - 12x^2 + 2$

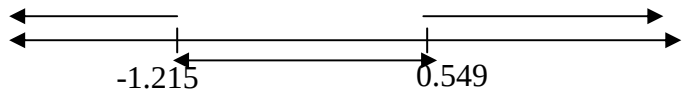
$$\begin{aligned} y' &= 12x^3 + 12x^2 - 24x \\ &= 12x(x^2 + x - 2) \\ &= 12x(x+2)(x-1) \end{aligned}$$



The sign analysis of y' is given below:

Interval	$12x$	$(x+2)$	$(x-1)$	y'	Conclusion
$x < -2$	- ve	- ve	- ve	-ve	y is decreasing on $(-\infty, -2]$
$-2 < x < 0$	- ve	+ ve	- ve	+ ve	y is increasing on $[-2, 0]$
$0 < x < 1$	+ ve	+ve	- ve	-ve	y is decreasing on $[0, 1]$
$x > 1$	+ ve	+ ve	+ ve	+ ve	y is increasing on $[1, \infty)$

$$\begin{aligned} y'' &= 36x^2 + 24x - 24 \\ &= 12(3x^2 + 2x - 2) \\ &= 12\left(x - \frac{-1+\sqrt{7}}{3}\right)\left(x - \frac{-1-\sqrt{7}}{3}\right) \\ &= 12(x-0.549)(x+1.215) \end{aligned}$$



The sign analysis of y'' is given below:

Interval	$(x-0.549)$	$(x+1.215)$	y''	Conclusion
$x < -1.215$	- ve	- ve	+ ve	y is concave up on $(-\infty, -1.215)$
$-1.215 < x < 0.549$	- ve	+ ve	- ve	y is concave down on $(-1.215, 0.549)$
$x > 0.549$	+ ve	+ve	+ve	y is concave up on $(0.549, \infty)$

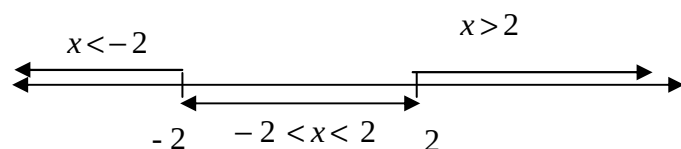
From the above sign analysis of y'' we see that $x = -1.215, 0.549$ are two inflection points of the given function.

Problem-2: Find the critical points of $f(x) = 5 + 12x - x^3$. Identify the intervals on which y is increasing and decreasing, concave up and concave down. Also find the inflection points, if any, of the above function.

Solution:

Here $f(x) = 5 + 12x - x^3$

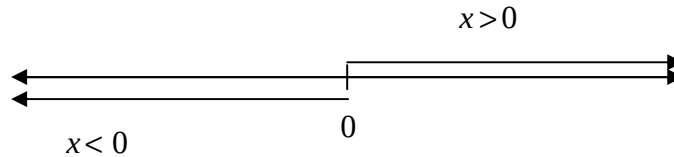
$$\begin{aligned} f'(x) &= 12 - 3x^2 \\ &= 3(4 - x^2) \\ &= 3(2+x)(2-x) \end{aligned}$$



The sign analysis of $f'(x)$ is given below:

Interval	$(x+2)$	$(2-x)$	$f'(x)$	Conclusion
$x < -2$	- ve	+ ve	- ve	$f(x)$ is decreasing on $(-\infty, -2]$
$-2 < x < 2$	+ ve	+ ve	+ ve	$f(x)$ is increasing on $[-2, 2]$
$x > 2$	+ ve	- ve	- ve	$f(x)$ is decreasing on $[2, \infty)$

$$f''(x) = -6x$$



The sign analysis of $f''(x)$ is given below:

Interval	$f''(x)$	Conclusion
$x < 0$	+ ve	$f(x)$ is concave up on $(-\infty, 0)$
$x > 0$	- ve	$f(x)$ is concave down on $(0, \infty)$

From the above sign analysis of $f''(x)$ we see that $x = 0$ is an inflection point of the given function.